

Kernel Engineering

From silicon to speculative decoding – GPU kernels for modern LLMs.

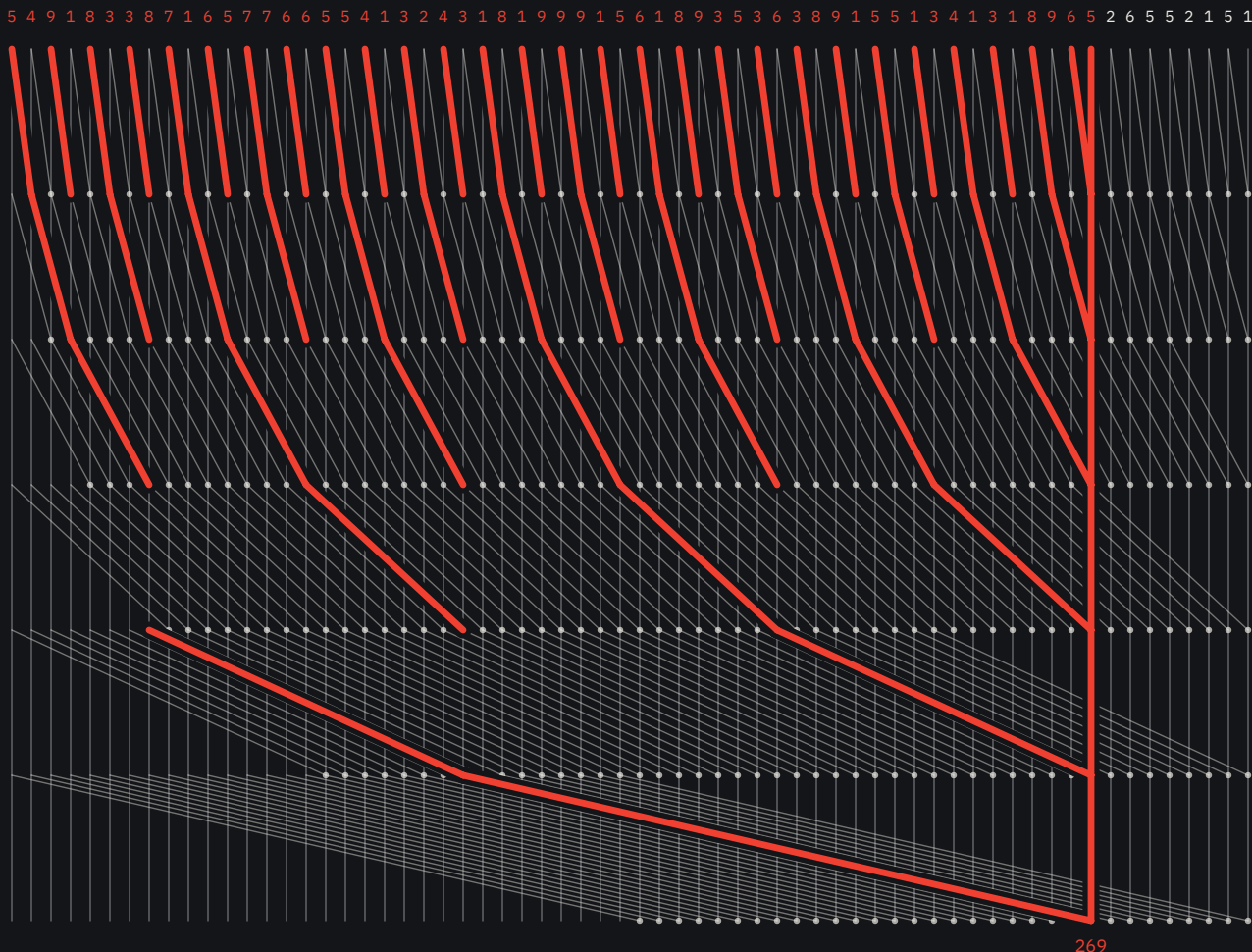


FIG. 0 A Kogge-Stone prefix sum, the pattern inside GPU scan kernels: 64 threads, 6 steps, 321 additions. At step k each thread adds the partial sum 2^k places to its left. In red: the additions thread 55 depends on, gathering 56 of the digits above into 269.